QUANTUM ERROR CORRECTION (QEC) — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 17, 2026

Diagram-based questions on Quantum Error Correction (QEC)

Question 1

10.0 points

Analyze the stabilizer code in the diagram below and explain how the code detects and corrects a single bit-flip error. Identify the role of each stabilizer and how they contribute to error detection.

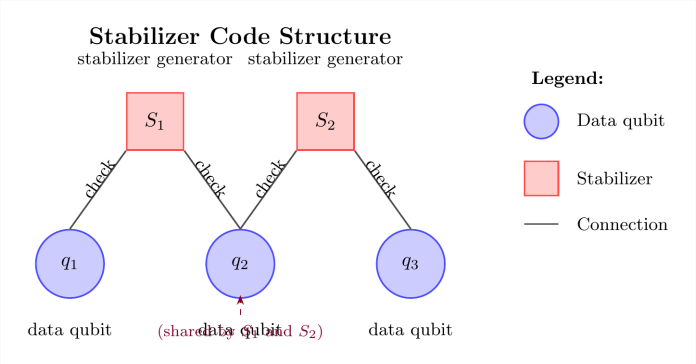

Fig. 1. Diagram of a simple stabilizer code illustrating error detection using syndromes. The diagram should show three data qubits and two stabilizer generators. Each stabilizer generator is connected to two of the data qubits, forming a check that can detect a single bit-flip error. Label the qubits as q1, q2, q3 and the stabilizers as S1, S2. Indicate the connections between stabilizers and qubits with lines, and show how a bit-flip error on one qubit affects the stabilizers' syndromes. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Hamiltonian The diagram will FAIL review if any of these appear.

Question 2

10.0 points

Given the circuit shown below of the 3-qubit bit-flip code, describe the process of detecting a single bit-flip error, including the logic involved in the syndrome measurement circuit.

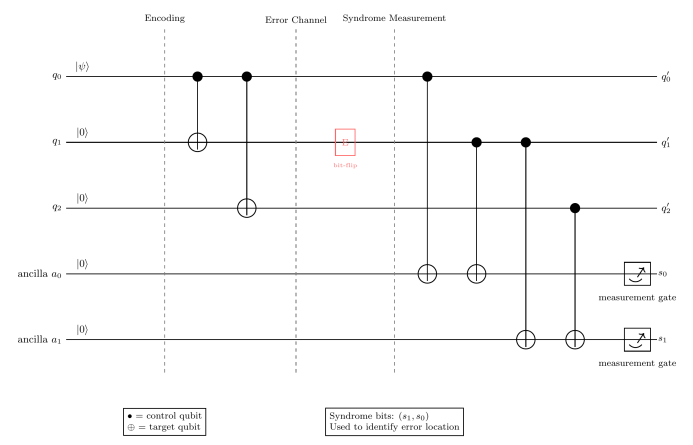

Fig. 2. 3-qubit bit-flip code circuit with syndrome measurement subcircuit. The circuit includes three qubit lines labeled q0, q1, and q2. The encoding section uses CNOT gates to entangle the qubits. The syndrome measurement subcircuit includes additional CNOT gates and measurement symbols to detect a single bit-flip error. The circuit should clearly show the flow from encoding to error detection, with labeled qubit lines and gates. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • CNOT The diagram will FAIL review if any of these appear.

Question 3

10.0 points

Examine the surface code lattice in the diagram below, and explain how logical qubits are formed and manipulated within the code.

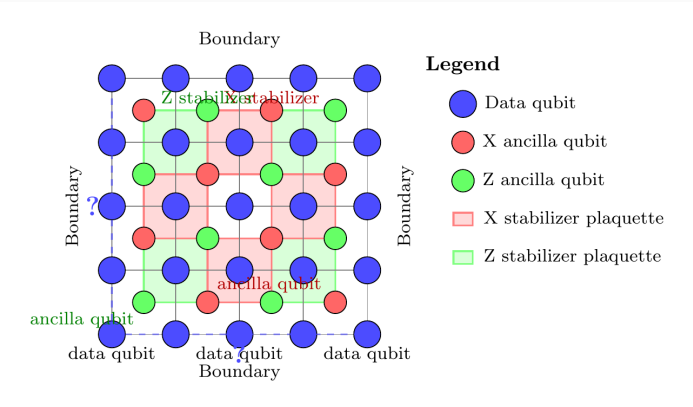

Fig. 3. Surface code lattice diagram showing a grid of physical qubits. Each node represents a physical qubit, arranged in a 2D lattice. Highlight the logical qubits by marking specific paths or clusters of physical qubits. Include labels for data qubits and ancilla qubits, and indicate logical operations such as X and Z stabilizers. Use a clear and organized layout to represent the surface code structure, emphasizing the connections and interactions between qubits. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • logical operator The diagram will FAIL review if any of these appear.

Question 4

10.0 points

Interpret the fault-tolerant gate teleportation in the diagram below, and describe the procedure for teleporting a quantum gate in a fault-tolerant quantum computing system.

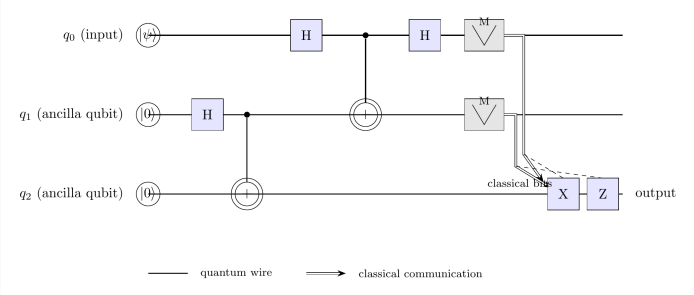

Fig. 4. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Initially; The diagram labels the distinct stages of the procedure ("Entanglement", "Gate Application", "Measurement", "Correction"), which directly answers the question asking the student to describe the procedure.. Remove the labels "Entanglement", "Gate Application", "Measurement", and "Correction". Keep all other labels and the overall circuit structure. The diagram should show the quantum circuit for gate teleportation, including qubits q0, q1, and ancilla, H and CNOT gates, measurement gates, and classical feedback for correction, but without explicitly labeling the procedural steps.

Question 5 10.0 points

Analyze the threshold theorem in the diagram below, and explain how it relates the error rates of physical qubits to logical qubit error correction thresholds.

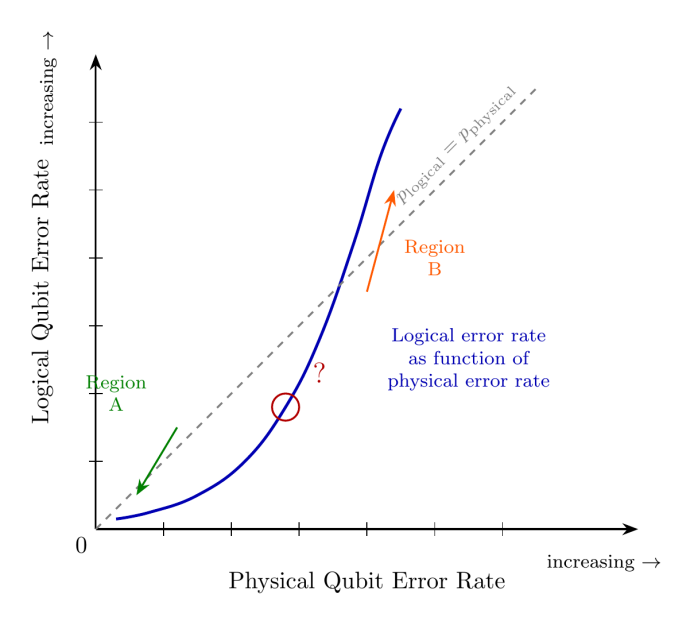

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: threshold; The diagram explicitly labels the regions as "Error correction effective" and "Error correction ineffective", which is the core conclusion the student is supposed to derive from analyzing the graph's shape.. A diagram illustrating the threshold theorem in quantum computing. The diagram should show a graph with error rates on the x-axis and logical qubit reliability on the y-axis. Include a curve that represents the threshold theorem, indicating a critical threshold point where error correction becomes effective. Label the axes as 'Physical Qubit Error Rate' and 'Logical Qubit Error Correction'. Highlight the threshold point where the error rate drops below the threshold, improving logical qubit reliability. Add annotations to explain the relationship between physical and logical qubit error rates in the context of the threshold theorem. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • threshold The diagram will FAIL review if any of these appear. IMPORTANT correction needed: The diagram explicitly labels the regions as "Error correction effective" and "Error correction ineffective", which is the core conclusion the student is supposed to derive from analyzing the graph's shape.. Specifically: The labels "Error correction effective", "Error correction ineffective", "Logical error decreases", and ". Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.